```
milos@komp:~$ docker --version
Docker version 29.3.0, build 5927d80
milos@komp:~$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
milos@komp:~$
```

```
milos@komp:~$ docker login

USING WEB-BASED LOGIN

i Info → To sign in with credentials on the command line, use 'docker login
-u <username>'


Your one-time device confirmation code is: WGFM-XCSC
Press ENTER to open your browser or submit your device code here: https://lo
gin.docker.com/activate

Waiting for authentication in the browser…

WARNING! Your credentials are stored unencrypted in '/home/milos/.docker/con
fig.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
milos@komp:~$
```

```
milos@komp:~$ docker pull hello-world
Using default tag: latest
latest: Pulling from library/hello-world
17eec7bbc9d7: Pull complete
ea52d2000f90: Download complete
Digest: sha256:85404b3c53951c3ff5d40de0972b1bb21fafa2e8daa235355baf44f33db9d
bdd
Status: Downloaded newer image for hello-world:latest
docker.io/library/hello-world:latest
milos@komp:~$ docker run hello world
Unable to find image 'hello:latest' locally
docker: Error response from daemon: pull access denied for hello, repository
 does not exist or may require 'docker login'

Run 'docker run --help' for more information
milos@komp:~$ docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent
it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/
```

```
milos@komp:~$ docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
01d7766a2e4a: Pull complete
fd8cda969ed2: Download complete
Digest: sha256:d1e2e92c075e5ca139d51a140fff46f84315c0fdce203eab2807c7e495eff
4f9
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest
milos@komp:~$ docker run ubuntu
milos@komp:~$ docker run ubuntu:latest cat /etc/os-release
PRETTY_NAME="Ubuntu 24.04.4 LTS"
NAME="Ubuntu"
VERSION_ID="24.04"
VERSION="24.04.4 LTS (Noble Numbat)"
VERSION_CODENAME=noble
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-
policy"
UBUNTU_CODENAME=noble
LOGO=ubuntu-logo
milos@komp:~$
```

```
milos@komp:~$ docker pull fedora:latest
latest: Pulling from library/fedora
5bc90b3315da: Pull complete
6fc39cdd7940: Download complete
c79496fc4cc5: Download complete
Digest: sha256:781b7642e8bf256e9cf75d2aa58d86f5cc695fd2df113517614e181a5eee9
138
Status: Downloaded newer image for fedora:latest
docker.io/library/fedora:latest
milos@komp:~$ docker run fedora:latest cat /etc/fedora-release
Fedora release 43 (Forty Three)
```

```
milos@komp:~$ docker images
                                                  i Info →   U  In Use
IMAGE                 ID             DISK USAGE   CONTENT SIZE   EXTRA
fedora:latest         781b7642e8bf        271MB         70.7MB    U
hello-world:latest    85404b3c5395       25.9kB         9.52kB    U
ubuntu:latest         d1e2e92c075e        119MB         31.7MB    U
milos@komp:~$
```

```
milos@komp:~$ docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
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To try something more ambitious, you can run an Ubuntu container with:
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For more examples and ideas, visit:
 https://docs.docker.com/get-started/

milos@komp:~$ echo $?
0
```

```
milos@komp:~$ docker run ubuntu:latest cat /etc/os-release
PRETTY_NAME="Ubuntu 24.04.4 LTS"
NAME="Ubuntu"
VERSION_ID="24.04"
VERSION="24.04.4 LTS (Noble Numbat)"
VERSION_CODENAME=noble
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-
policy"
UBUNTU_CODENAME=noble
LOGO=ubuntu-logo
milos@komp:~$ echo $?
0
milos@komp:~$ docker run fedora:latest cat /etc/fedora-release
Fedora release 43 (Forty Three)
milos@komp:~$ echo $?
0
```

```
milos@komp:~$ docker run --rm busybox echo "Kontener busybox uruchomiony"
Unable to find image 'busybox:latest' locally
latest: Pulling from library/busybox
61dfb50712f5: Pull complete
96cfb76e59bd: Download complete
Digest: sha256:b3255e7dfbcd10cb367af0d409747d511aeb66dfac98cf30e97e87e4207dd
76f
Status: Downloaded newer image for busybox:latest
Kontener busybox uruchomiony
```

```
milos@komp:~$ docker run -it --rm busybox sh
/ # busybox --help
BusyBox v1.37.0 (2024-09-26 21:31:42 UTC) multi-call binary.
BusyBox is copyrighted by many authors between 1998-2015.
Licensed under GPLv2. See source distribution for detailed
copyright notices.

Usage: busybox [function [arguments]...]
   or: busybox --list[-full]
   or: busybox --show SCRIPT
   or: busybox --install [-s] [DIR]
   or: function [arguments]...

        BusyBox is a multi-call binary that combines many common Unix
        utilities into a single executable.  Most people will create a
        link to busybox for each function they wish to use and BusyBox
        will act like whatever it was invoked as.

Currently defined functions:
        [, [[, acpid, add-shell, addgroup, adduser, adjtimex, ar, arch,
        arp, arping, ascii, ash, awk, base32, base64, basename, bc, beep,
        blkdiscard, blkid, blockdev, bootchartd, brctl, bunzip2, bzcat,
        bzip2, cal, cat, chat, chattr, chgrp, chmod, chown, chpasswd,
        chpst, chroot, chrt, chvt, cksum, clear, cmp, comm, conspy, cp
```

```
milos@komp:~$ docker run -it --name my-fedora fedora:latest bash
[root@ab64bfd27b7e /]# ps aux
bash: ps: command not found
[root@ab64bfd27b7e /]# cat /proc/1/cmdline
bash[root@ab64bfd27b7e echo "PID1: $(cat /proc/1/status | grep ^Pid)"id)"
PID1: Pid:      1
[root@ab64bfd27b7e /]# dnf update
Updating and loading repositories:
 Fedora 43 openh264 (From Cisco) -  100% |   1.4 KiB/s |   5.8 KiB |  00m04s
 Fedora 43 - x86_64 - Updates       100% |   2.0 MiB/s |  12.5 MiB |  00m06s
 Fedora 43 - x86_64                 100% |   1.7 MiB/s |  35.4 MiB |  00m20s
Repositories loaded.
Package                                 Arch   Version               Reposi
      Size
Upgrading:
 ca-certificates                        noarch 2025.2.80_v9.0.304-1. update
   2.7 MiB
   replacing ca-certificates            noarch 2025.2.80_v9.0.304-1. 3b3d0b
   2.7 MiB
 coreutils                              x86_64 9.7-8.fc43            update
   5.4 MiB
   replacing coreutils                  x86_64 9.7-7.fc43            a899a9
   5.4 MiB
 coreutils-common                       x86_64 9.7-8.fc43            update
  11.3 MiB
```

```
FROM ubuntu:latest

ENV DEBIAN_FRONTEND=noninteractive

RUN apt-get update && \
    apt-get upgrade -y && \
    apt-get install -y --no-install-recommends git ca-certificates && \
    update-ca-certificates --fresh && \
    apt-get clean && \
    rm -rf /var/lib/apt/lists/*

WORKDIR /repo

RUN git clone https://github.com/InzynieriaOprogramowaniaAGH/MDO2026_ITE.gi>

CMD ["/bin/bash"]
```

```
milos@komp:~/MDO2026_ITE/grupa2/MF419850/sprawozdanie2$ docker build -t mdo-
repo-image .
[+] Building 170.7s (8/8) FINISHED                          docker:default
 => [internal] load build definition from Dockerfile                  0.1s
 => => transferring dockerfile: 436B                                  0.0s
 => [internal] load metadata for docker.io/library/ubuntu:latest      0.3s
 => [internal] load .dockerignore                                     0.1s
 => => transferring context: 2B                                       0.0s
 => CACHED [1/4] FROM docker.io/library/ubuntu:latest@sha256:d1e2e92c  0.4s
 => => resolve docker.io/library/ubuntu:latest@sha256:d1e2e92c075e5ca  0.3s
 => [2/4] RUN apt-get update &&     apt-get upgrade -y &&     apt-ge  89.2s
 => [3/4] WORKDIR /repo                                               0.8s
 => [4/4] RUN git clone https://github.com/InzynieriaOprogramowaniaA  46.1s
 => exporting to image                                               32.6s
 => => exporting layers                                              27.3s
 => => exporting manifest sha256:9ce781a1a68bb74916c295cfa385aa212d94  0.1s
 => => exporting config sha256:f7e798877f23ff0fbbaef7ca0f03de88e7c3f8  0.1s
 => => exporting attestation manifest sha256:f91d19ad42010f72aef49a7b  0.1s
 => => exporting manifest list sha256:05e059533689b9ab7c1efead605b2fb  0.1s
 => => naming to docker.io/library/mdo-repo-image:latest              0.0s
 => => unpacking to docker.io/library/mdo-repo-image:latest           4.6s
milos@komp:~/MDO2026_ITE/grupa2/MF419850/sprawozdanie2$ nano Dockerfile
```

```
milos@komp:~/MDO2026_ITE/grupa2/MF419850/sprawozdanie2$ docker run -it --nam
e mdo-repo-container mdo-repo-image
root@3f06d23a6d70:/repo# git --version
git version 2.43.0
root@3f06d23a6d70:/repo# ls -la /repo/
total 24
drwxr-xr-x 1 root root 4096 Mar 13 09:56 .
drwxr-xr-x 1 root root 4096 Mar 13 10:00 ..
drwxr-xr-x 8 root root 4096 Mar 13 09:56 .git
-rw-r--r-- 1 root root   45 Mar 13 09:56 README.md
drwxr-xr-x 2 root root 4096 Mar 13 09:56 READMEs
drwxr-xr-x 3 root root 4096 Mar 13 09:56 grupa4
root@3f06d23a6d70:/repo# cd /repo
root@3f06d23a6d70:/repo# git branch a
root@3f06d23a6d70:/repo# git branch -a
  a
* main
  remotes/origin/AD420339
  remotes/origin/AD_420339-NEW
  remotes/origin/AJ420697
  remotes/origin/AN420700
  remotes/origin/AP421868
  remotes/origin/BN419779
  remotes/origin/BW414729
  remotes/origin/BW419687
  remotes/origin/DM
  remotes/origin/DM421198
  remotes/origin/DP423171
  remotes/origin/DS419547
  remotes/origin/DW419755
  remotes/origin/FP422032
  remotes/origin/FT422048
  remotes/origin/GN421256
  remotes/origin/HEAD -> origin/main
```

```
milos@komp:~/MDO2026_ITE/grupa2/MF419850/sprawozdanie2$ docker ps -a
CONTAINER ID   IMAGE            COMMAND                  CREATED          ST
ATUS                            PORTS     NAMES
3f06d23a6d70   mdo-repo-image   "/bin/bash"              3 minutes ago    Ex
ited (130) About a minute ago             mdo-repo-container
ab64bfd27b7e   fedora:latest    "bash"                   31 minutes ago   Ex
ited (0) 25 minutes ago                   my-fedora
2807a603b74c   fedora:latest    "cat /etc/fedora-rel…"   38 minutes ago   Ex
ited (0) 38 minutes ago                   affectionate_wiles
81ab49a9de2d   ubuntu:latest    "cat /etc/os-release"    38 minutes ago   Ex
ited (0) 38 minutes ago                   vigorous_agnesi
90457a6dddbe   hello-world      "/hello"                 39 minutes ago   Ex
ited (0) 39 minutes ago                   heuristic_proskuriakova
2f420fac972e   fedora:latest    "cat /etc/fedora-rel…"   40 minutes ago   Ex
ited (0) 40 minutes ago                   gifted_chebyshev
1d26b93173f4   ubuntu:latest    "cat /etc/os-release"    42 minutes ago   Ex
ited (0) 42 minutes ago                   silly_driscoll
37f2641aa37a   ubuntu           "/bin/bash"              43 minutes ago   Ex
ited (0) 43 minutes ago                   stoic_germain
ef06933a3ed0   hello-world      "/hello"                 44 minutes ago   Ex
ited (0) 44 minutes ago                   eager_diffie
milos@komp:~/MDO2026_ITE/grupa2/MF419850/sprawozdanie2$ docker container pru
ne -f
Deleted Containers:
3f06d23a6d7063230db0c25e303b799c5415b48e57ec0b5a81cc89e9d72032de
ab64bfd27b7eb0c5cd55cc3b8bcb47580e9f063cccee9cf32cb09a917e5d9416
2807a603b74c8162bb08ddb10815381854ab739f7d3c43967a059bd7c4e10d20
81ab49a9de2da889967546036669cf19b829f03e2dd4d0302c7f838f40baf78a
90457a6dddbe1d93863daf017daca6bb4c776fc50b8beb0bb43b8c4420769141
2f420fac972e50a9526382a445a6db3cec1a3b3ca697988d99e0552a7b2cf2bf
1d26b93173f4601726c5e86cde1f4f726ed3259878021d6e1e72b3e3f277c7b3
37f2641aa37a40bc69f903ad2491b5b212fe2ef20bd5799dbe0277ca583baecf
ef06933a3ed0940e4d3ba920abfa55576e88a2c6b01733a8ca8a40dd26dcc8b4

Total reclaimed space: 161.5MB
```

```
milos@komp:~/MDO2026_ITE/grupa2/MF419850/sprawozdanie2$ docker images
                                                    i Info →   U  In Use
IMAGE                     ID             DISK USAGE   CONTENT SIZE   EXTRA
busybox:latest            b3255e7dfbcd       6.77MB         2.22MB
fedora:latest             781b7642e8bf        271MB         70.7MB
hello-world:latest        85404b3c5395       25.9kB         9.52kB
mdo-repo-image:latest     05e059533689        405MB          141MB
ubuntu:latest             d1e2e92c075e        119MB         31.7MB
milos@komp:~/MDO2026_ITE/grupa2/MF419850/sprawozdanie2$ docker image prune -a -f
Deleted Images:
untagged: ubuntu:latest
deleted: sha256:579dfd6f4570917f7b45bffcabe073c57b49fc741d28371bae9f8b7aa20196b6
untagged: busybox:latest
deleted: sha256:b3255e7dfbcd10cb367af0d409747d511aeb66dfac98cf30e97e87e4207dd76f
deleted: sha256:70ce0a747f09cd7c09c2d6eaeab69d60adb0398f569296e8c0e844599388ebd6
deleted: sha256:afa90197716ae01be9facc62cdd0d029caaacd89d7aceea84b828817f7251b94
untagged: fedora:latest
deleted: sha256:781b7642e8bf256e9cf75d2aa58d86f5cc695fd2df113517614e181a5eee9138
deleted: sha256:abc510d524d2eec700ca4a51d5534f038f38ed8bd3ea3951c7b3c84718587093
deleted: sha256:4fff2f47f26392f76ffbdb566c6a14059d719e37914b8f072bd9fca7ad908c66
untagged: hello-world:latest
deleted: sha256:85404b3c53951c3ff5d40de0972b1bb21fafa2e8daa235355baf44f33db9dbdd
deleted: sha256:2771e37a12b7bcb2902456ecf3f29bf9ee11ec348e66e8eb322d9780ad7fc2df
deleted: sha256:6b75187531c5e9b6a85c8946d5d82e4ef3801e051fbff338f382f3edfa60e3d2
untagged: mdo-repo-image:latest
deleted: sha256:05e059533689b9ab7c1efead605b2fbbe579fc2db831fdc836569d897333ac63
deleted: sha256:9ce781a1a68bb74916c295cfa385aa212d94cb46a698fd7f53a3c9a5a85ab375
deleted: sha256:f91d19ad42010f72aef49a7bcc64a444b1c167c20b9dfc2112612435f38cda1f

Total reclaimed space: 74.85MB
```